Todo List Web Application

1. Project Title and Objective

Title: Todo List Web Application

Objective: To develop a full-stack web application that allows users to manage daily tasks efficiently. The application enables users to create, update, complete, and delete tasks with deadlines, while storing data securely in a database.

2. Project Description

This project is a full-stack Todo List application built using Node.js, MongoDB, and frontend technologies. Users can sign up, log in, and access a dashboard where they can manage tasks. Each task includes a title and a target date/time. The system automatically marks tasks as 'Pending' if the deadline is crossed and allows marking tasks as completed.

The application demonstrates CRUD operations, authentication, and real-time UI updates.

3. Technologies Used

Frontend:
- HTML
- CSS
- JavaScript

Backend:
- Node.js
- Express.js

Database:
- MongoDB (MongoDB Compass)

Other Tools:
- JWT (Authentication)
- Fetch API

4. Setup Instructions

1. Install Node.js and MongoDB.
2. Start MongoDB service.
3. Open backend folder in terminal.

4. Run: npm install
5. Start server using: node server.js
6. Open frontend files (login.html) in browser.
7. Signup and login to access dashboard.
8. Add tasks and verify data in MongoDB Compass.

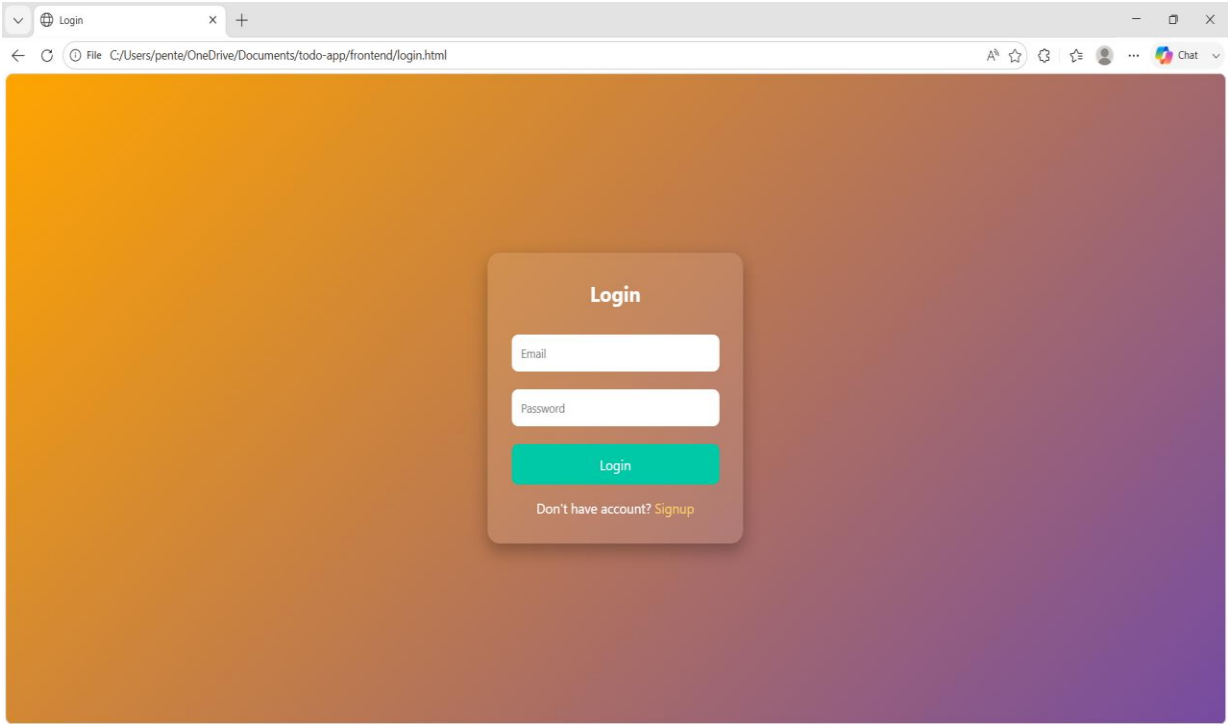
5. API Endpoints Explanation

POST /signup - Register a new user
POST /login - Authenticate user and return token
POST /addTask - Add new task (requires token)
GET /tasks - Get all tasks for logged-in user
PUT /complete/:id - Mark task as completed
DELETE /delete/:id - Delete task

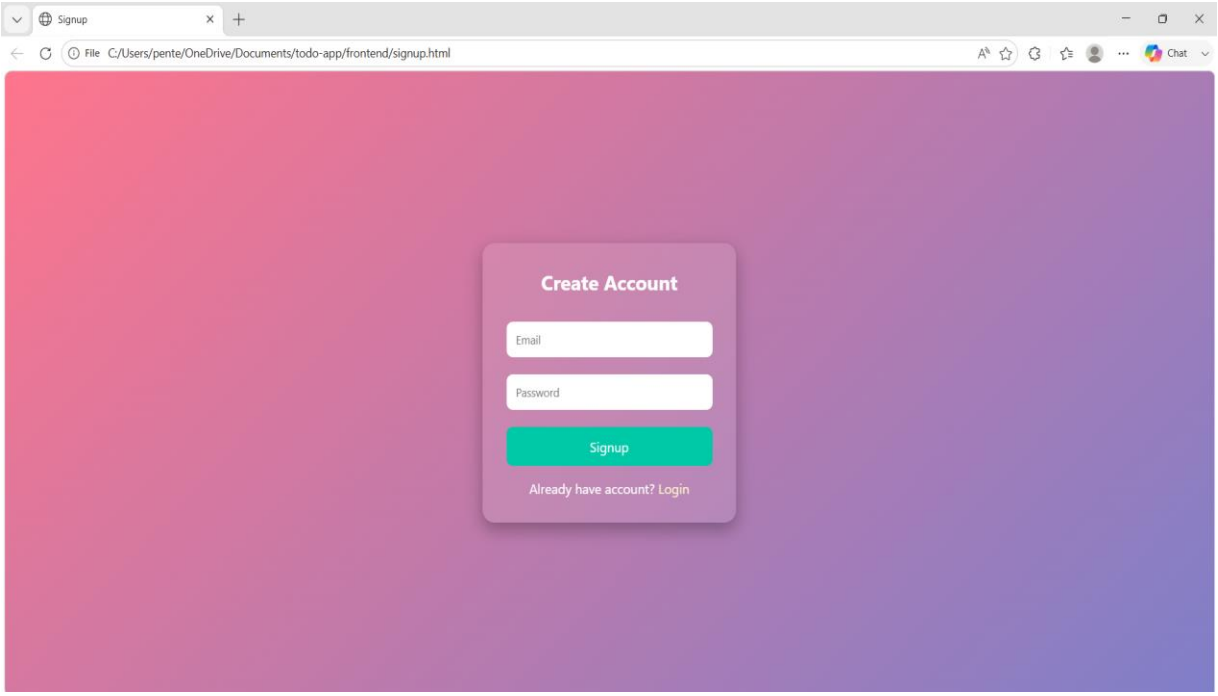
6. Screenshots of the Project

1.UI SCREENSHOTS:

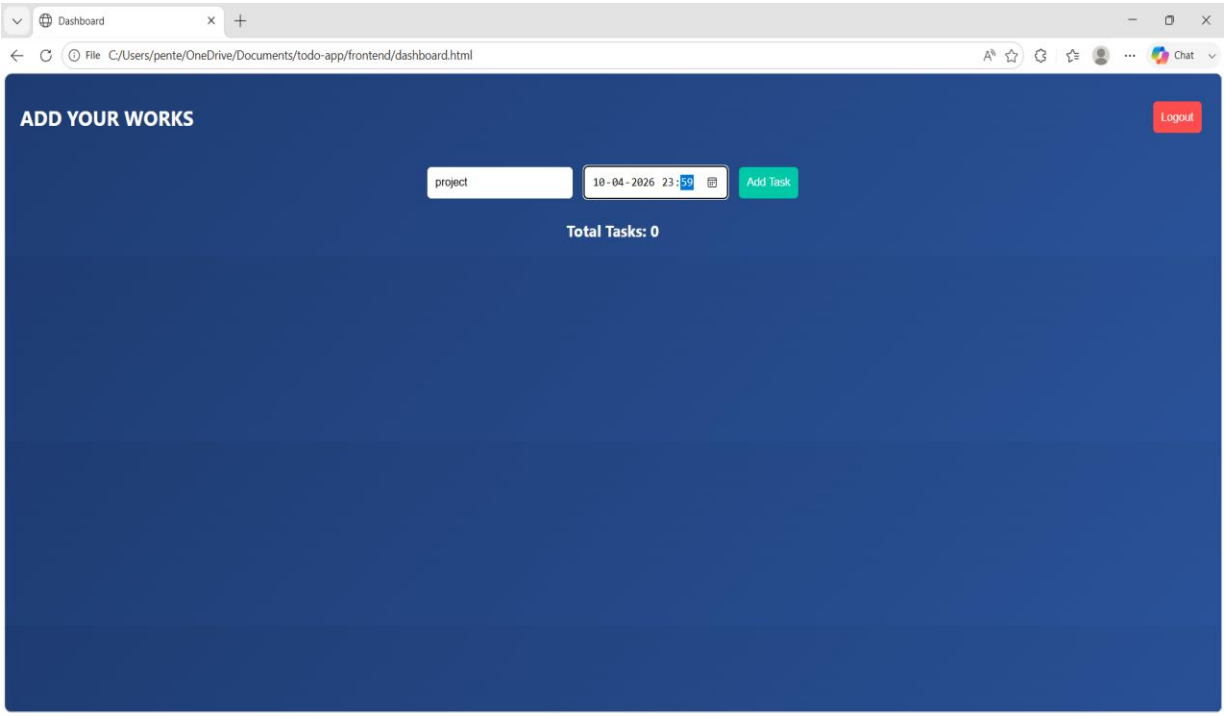
Login Page:



Signup Page:

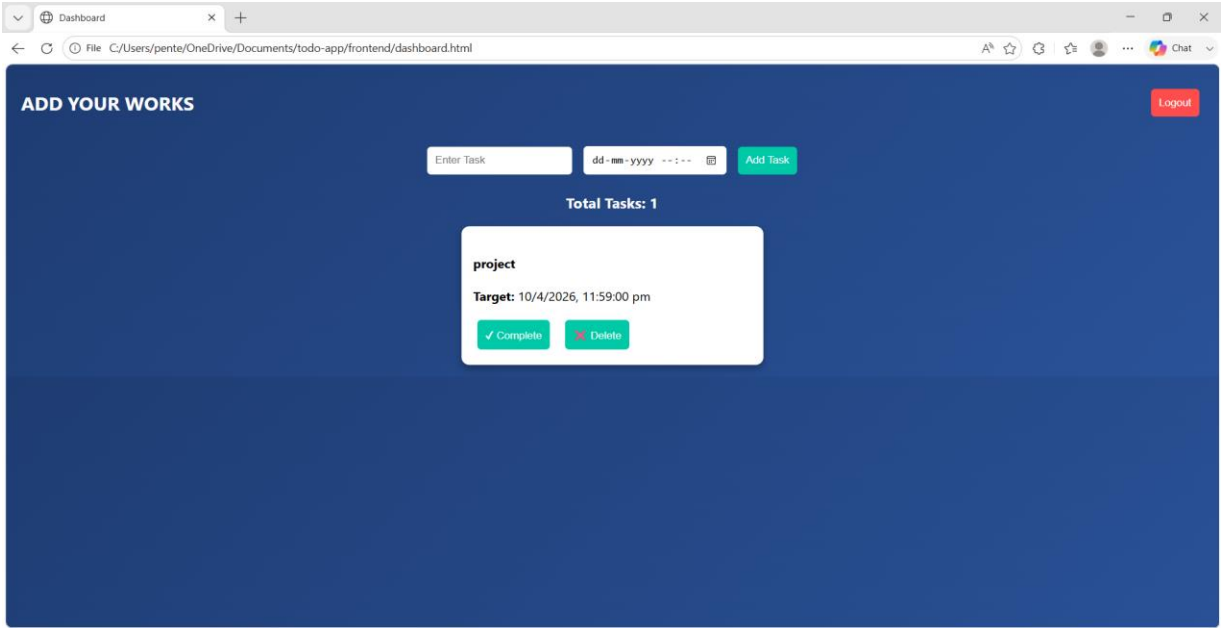


Dashboard:

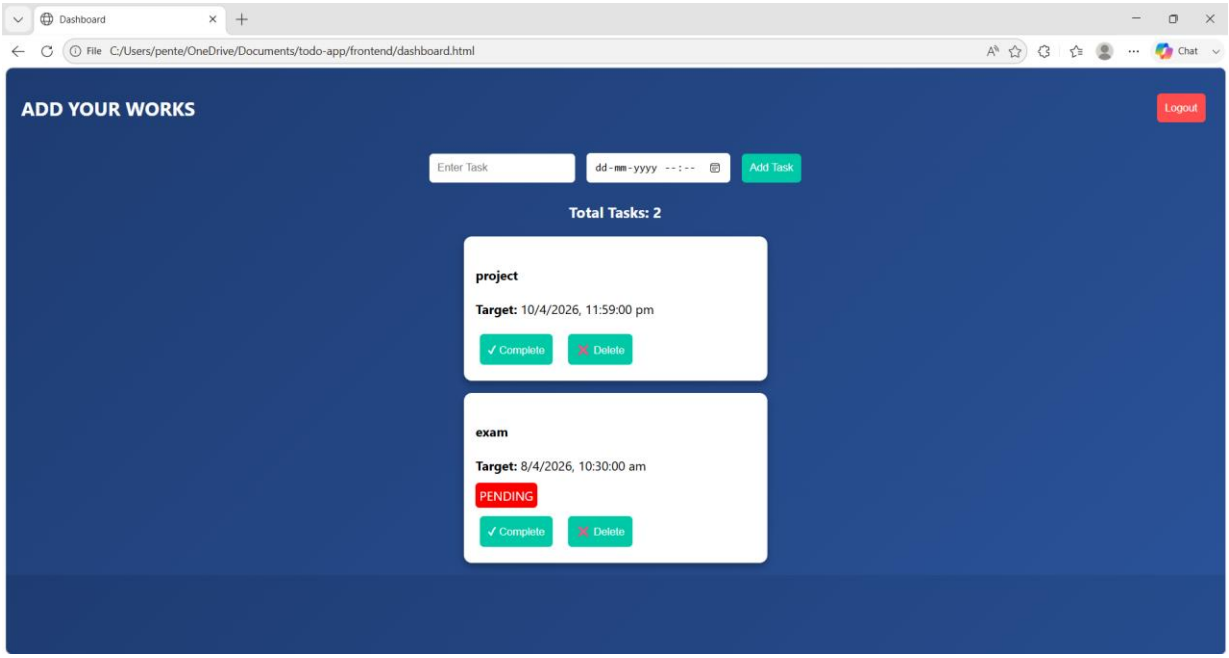


2.TASK OPERATIONS SCREENSHOTS:

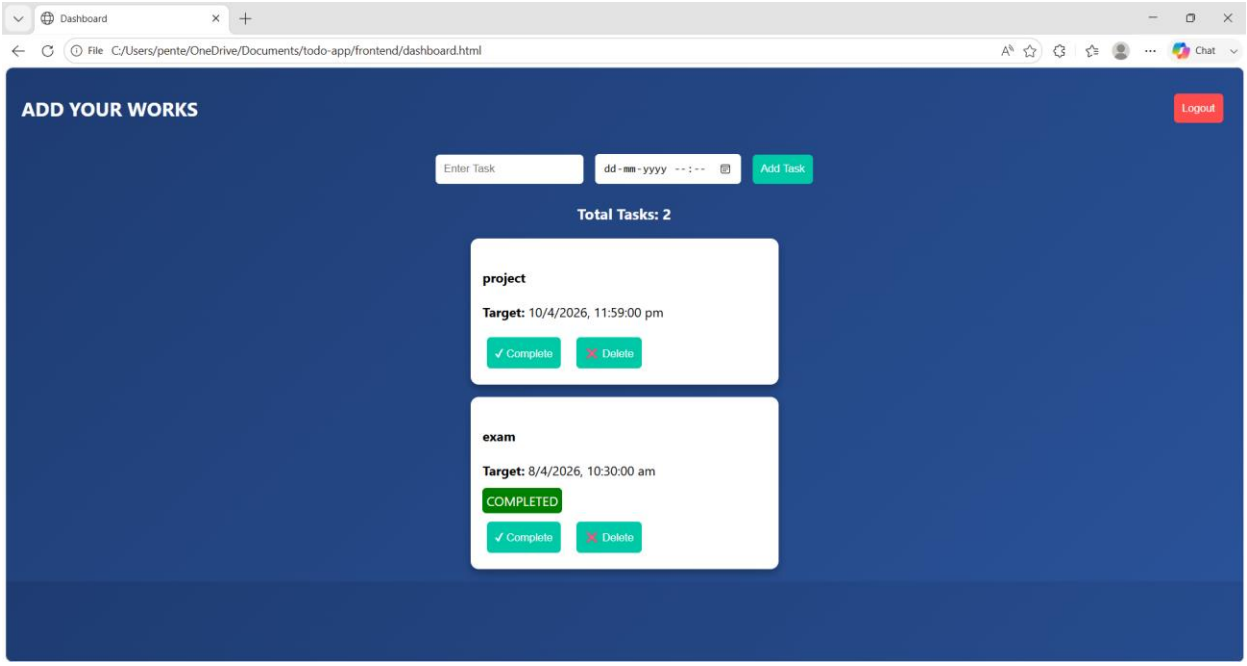
Task ADDED:



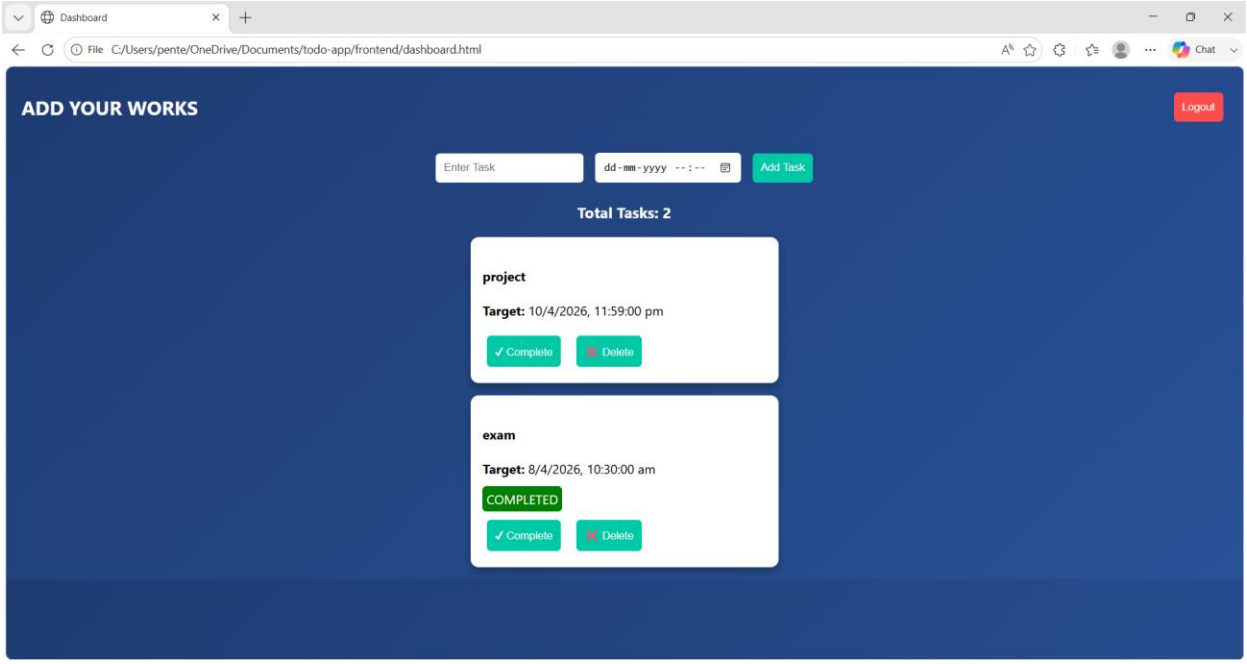
Task UPDATED:



Task COMPLETED:

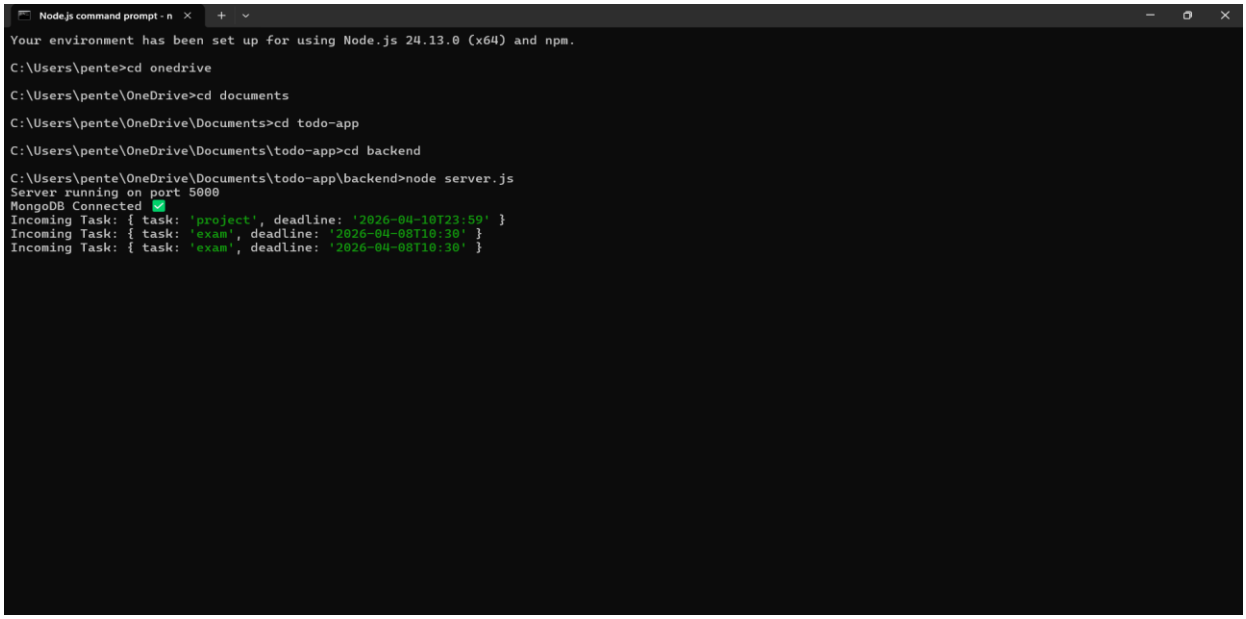


Task DELETED:

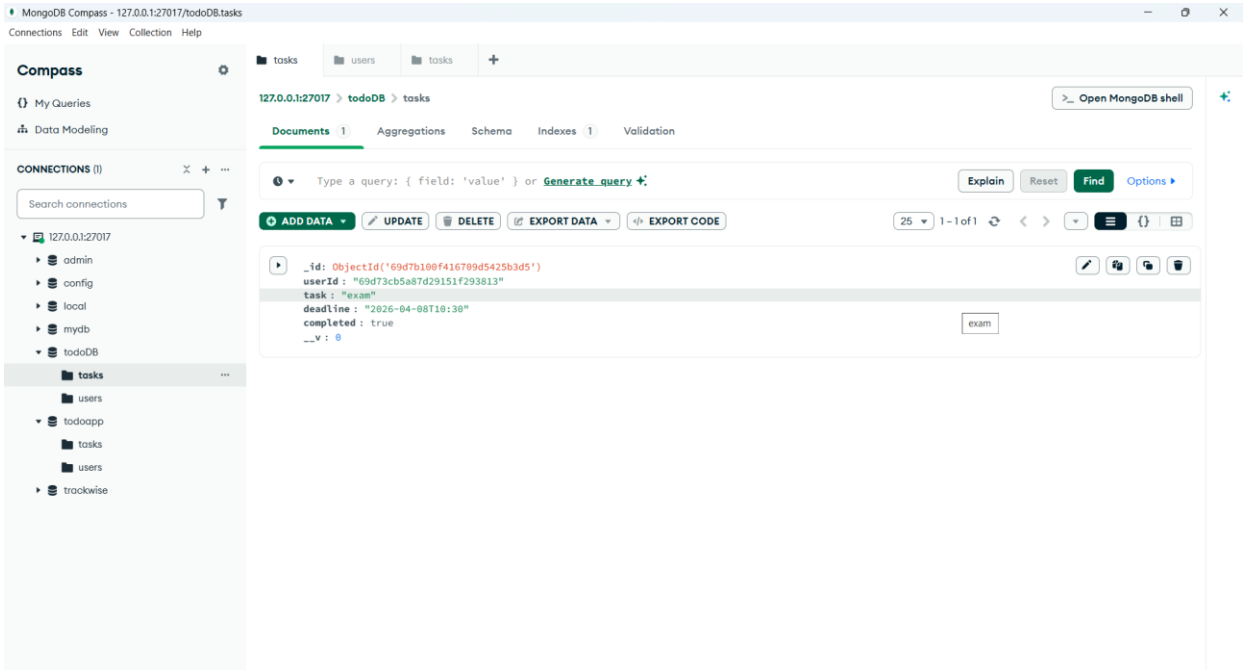


3.BACKEND SCREENSHOTS:

Server Running:



MongoDB Data View:



7. Challenges Faced

- Connecting MongoDB with Node.js
- Handling authentication using JWT
- Managing frontend-backend communication
- Debugging API errors
- Designing responsive UI

These challenges were resolved through debugging, testing APIs, and improving code structure.